Field Officer Offline Data Collection App

A companion application to PMCC (Prime Minister Contact Center)

BRD + PRD — for an external/separate app development team

How this app relates to PMCC

PMCC is a web-based complaint management system (Laravel + React) used by citizens, Focal Persons, Directors, Admin, and Super Admin. One workflow within it — a Focal Person dispatching a **field officer** to physically visit a complaint site — assumes the officer can access the web portal live. In practice, many parts of Azad Jammu & Kashmir have unreliable or absent mobile data/internet connectivity, particularly the mountainous and rural areas where field visits are often most needed. A web app that requires a live connection is unusable exactly where and when it matters most.

This document scopes a **separate, purpose-built mobile app** to solve that specific problem — not a feature bolted onto the main PMCC web system, but its own small application that talks to PMCC's backend when connectivity allows, and works fully offline when it doesn't.

This app does not replace any part of PMCC. Focal Persons, Directors, Admin, and Super Admin continue using the web portals exactly as already built. Only the field officer's specific task — visiting a site and recording findings — moves to this dedicated tool.

Part 1 — Business Requirements Document (BRD)

Background

PMCC routes citizen complaints to government departments, where a Focal Person may need a field officer to physically inspect a site (e.g. a damaged road, a school facility issue, an infrastructure complaint) before the complaint can be resolved. AJK's terrain and connectivity infrastructure mean an officer dispatched to many locations cannot reliably access a live web application while there.

Business Objective

Ensure that a field visit's findings — notes, photographs, and location — are reliably captured at the time and place the visit actually happens, regardless of network availability, and are automatically delivered to PMCC once the officer's device regains connectivity, without requiring the officer to remember to "re-enter" anything later or risk losing what they recorded.

Scope

In scope:

- A mobile app for field officers to view their assigned visits and record findings, usable with no internet connection at the point of use.
- Automatic, reliable synchronization of that recorded data back into PMCC's existing database once connectivity is available.
- Authentication using the same field officer accounts already provisioned inside PMCC (Module 2/4's `users` table, role `field_officer`).

Out of scope:

- Any Focal Person, Director, Admin, or Super Admin functionality — those roles continue using the existing PMCC web portals.
- Creating or classifying complaints — this app only handles the field-visit step of an already-existing, already-assigned complaint.
- Any offline capability for roles other than field officers.

Stakeholders

- **Field officers** — the app's primary and only users.
- **Focal Persons** — assign visits (via the existing PMCC web portal) and consume the synced results once they arrive.
- **PITB (Super Admin)** — technical authority; will need to provision API access/credentials for this app to talk to PMCC's backend.
- **Citizens** — indirect beneficiaries; the point of this app is that their complaint doesn't stall just because the assigned officer was somewhere without signal.

High-Level Business Requirements

1. A field officer can see the visits assigned to them without needing a live connection at the moment they check, as long as their device synced at some point before going offline.

2. A field officer can record what they find at a site — written notes, photos, and ideally the location — with zero internet connection required during the visit itself.
3. Nothing a field officer records offline is ever silently lost; it queues on the device and is delivered to PMCC as soon as a connection becomes available, without the officer needing to do anything manual to trigger it.
4. A field officer can always tell, at a glance, which of their recorded visits have successfully reached PMCC and which are still waiting to sync.
5. This app authenticates against the same accounts already set up in PMCC — no separate account system, no separate password for the officer to manage.

Assumptions & Constraints

- **Device platform is not yet decided.** This BRD doesn't mandate Android, iOS, or a specific technology — that's a decision for the app development team, informed by what devices AJK field officers actually carry (see Open Questions).
- PMCC's backend (Laravel) will need to expose a documented API for this app to consume — this doesn't exist yet and is a dependency this project creates for the main PMCC build, detailed in Part 2.
- This app is intentionally small and single-purpose. Resist scope creep toward becoming a general-purpose field-work app beyond what PMCC's field-visit workflow actually needs.

Success Criteria

- Field visit findings are captured and eventually delivered to PMCC for **100% of assigned visits**, even those conducted with zero connectivity throughout.
- Time between "visit conducted" and "findings visible to the Focal Person in PMCC" shrinks to whenever the officer's device next has any connectivity — not "whenever the officer manually remembers to log back into a web portal."

Part 2 — Product Requirements Document (PRD)

Problem Statement

A field officer assigned a visit in a low- or no-connectivity area currently has no reliable way to receive full visit details or record findings through PMCC at the time of the visit. Today's alternative — writing notes on paper or using their phone's ordinary camera, then manually re-entering everything into the web portal later — is error-prone, easy to forget, and loses structure (photos disconnected from the specific complaint, no consistent record format).

Goals

1. A field officer can view an assigned visit's details (complaint summary, location, complainant contact) entirely offline, as long as it synced to their device at some earlier point.
2. A field officer can capture structured findings — notes, photos, GPS location, timestamp — with the device fully offline, with nothing lost.
3. Recorded findings sync automatically and reliably to PMCC's database the moment connectivity returns, with no duplicate or corrupted submissions even if connectivity drops and returns multiple times mid-sync.
4. A field officer always has a clear, simple visual indicator of what's synced vs. still pending.
5. Logging into this app uses the exact same credentials as the officer's existing PMCC `field_officer` account.

Non-Goals

- **Any functionality for roles other than field officer.** Focal Persons, Directors, Admin, and Super Admin never use this app.
- **Complaint creation, classification, or resolution.** This app only records findings for a visit that already exists and was already assigned inside PMCC.
- **Replacing the Focal Person's Action/Resolution screen.** The FP still formally resolves the complaint in the PMCC web portal, using the synced findings from this app as input.
- **A generalized offline mode for the rest of PMCC.** This is intentionally scoped to the one workflow that genuinely needs it.

User Stories

- As a **field officer**, I want to see my assigned visits on my phone even without signal, so I'm not stuck if I lose connectivity between checking my assignments and arriving at the site.
- As a **field officer**, I want to write notes and take photos during a visit with no internet connection at all, so poor signal at the actual site never stops me from doing my job.
- As a **field officer**, I want my recorded findings to upload automatically once I'm back in range, so I don't have to remember to manually resend anything.
- As a **field officer**, I want to see clearly which of my visit records have and haven't reached the system, so I know if I need to keep my phone connected a bit longer before considering the job done.
- As a **field officer**, I want to log in with my existing PMCC account, so I don't have to manage a second set of credentials.

- As a **Focal Person**, I want a field officer's synced findings to appear automatically in the same Investigation Log I already use in PMCC, so I don't have to go looking in a separate system.

Requirements

P0 — Must-Have | Requirement | Acceptance Criteria | |---|---| | Offline-viewable assigned visits | Given the app has synced at least once, when the device has no connection, then the officer can still see their assigned visits' full details (complaint summary, location, complainant contact). | | Offline data capture | Given no connection, the officer can write notes, attach photos, and capture GPS location for a visit, and none of this data is lost if the app is closed or the device restarts before syncing. | | Automatic background sync | Given connectivity becomes available, then queued findings upload automatically without the officer needing to open a specific "sync" screen or take manual action. | | Sync status indicator | Every recorded visit finding shows a clear status: Pending Sync or Synced. | | Duplicate-safe sync | Given a sync is interrupted partway (connection drops mid-upload), when connectivity returns, then the record is not duplicated or corrupted on retry. | | Shared authentication with PMCC | The officer logs in using their existing PMCC `field_officer` account credentials; no separate account exists. | | Data maps to existing PMCC schema | A synced finding populates the same `complaint_investigations` fields the web portal already uses: `assigned_officer_id`, `visit_datetime`, `location`, `notes`, plus associated `complaint_attachments` rows for any photos. |

P1 — Nice-to-Have

- Push notification when a new visit is assigned, delivered whenever the device next has any connectivity (data or SMS-triggered, depending on final platform choice).
- A simple checklist/template for common visit types, so notes are more structured than free text alone.
- Ability to view (not just create) past visits the officer has already completed, offline.

P2 — Future Considerations

- Audio note capture as an alternative to typed notes (mirrors the voice-note idea raised and deferred for the main Citizen Portal — same transcription/AI-pipeline considerations would apply if ever built).
- Offline map caching, if GPS/mapping needs grow beyond a simple coordinate capture.

Technical Integration Requirements (dependency on the main PMCC backend)

This app cannot be built in isolation — it depends on new API endpoints that don't exist yet in PMCC's Laravel backend (which currently only serves Inertia web pages, not a general-purpose API). Whoever builds PMCC's backend needs to add, alongside the existing web routes:

- A **token-based authentication endpoint** (e.g. Laravel Sanctum) so this app can log a `field_officer` in and make authenticated requests, separate from the web portal's session-based login.
- A **"my assigned visits" endpoint**, scoped to the authenticated officer (matching the same `assigned_officer_id` scoping already enforced in the web portal), returning everything needed to render a visit offline.
- A **"submit visit findings" endpoint**, accepting notes, GPS coordinates, and photo uploads, writing to the same `complaint_investigations` and `complaint_attachments` tables the web portal already uses — so a Focal Person sees synced findings in exactly the same Investigation Log, with no separate data model to reconcile.

This API layer is new scope for the main PMCC build, not something this field app team can build themselves — it needs to be coordinated with whoever is developing PMCC's backend in Antigravity.

Success Metrics

- **Leading**: % of field visits with findings successfully synced within 24 hours of the visit taking place — the core metric this app exists to improve.
- **Leading**: 0 duplicate or corrupted records created from interrupted sync attempts, during testing.
- **Lagging**: reduction in field visits with no findings ever logged at all (today's likely failure mode with paper/camera-roll workarounds), once this app is in use.

Open Questions

- **What devices do AJK field officers actually carry?** This determines the platform choice (Android-only is the safer/cheaper default given typical government-issued or personally-owned device patterns in Pakistan, but should be confirmed rather than assumed before development starts).
- **Is GPS location capture actually needed**, or is the officer's typed/selected location description (already captured in the existing `complaint_investigations.location` field) sufficient? GPS adds real complexity

(permissions, accuracy in mountainous terrain, battery use) — worth confirming it's a genuine requirement rather than a nice-to-have before committing to it as P0.

- **Does the Focal Person also need any offline capability**, or is it truly only the field officer who works in disconnected areas? (This BRD assumes FPs have adequate connectivity at their department office, only the field visit itself happens somewhere disconnected — confirm this assumption holds.)
- **Who owns/hosts the API layer this app depends on** — is it built as part of the same Antigravity-driven PMCC project, or does the separate app development team need to coordinate a formal API contract with that team directly?

Handoff note for the app development team

This document is meant to be read on its own, without needing the full PMCC build history. The one thing worth requesting from the PMCC side before development starts: the exact current field names and validation rules for `complaint_investigations`, `complaint_attachments`, and the `field_officer` role on `users`, so this app's data model lines up exactly with what PMCC expects to receive — rather than guessing and reconciling mismatches later.